

Exploring reasons for Unclaimed Shifts and Revenue Loss

After carefully analyzing the dataset and its features, several key questions emerged in my mind. One of the first things I noticed was the high number of shift offers for some of the shifts, which led me to wonder why so many shifts remained unclaimed. Additionally, I questioned whether workers were willing to work on weekends, as I personally wouldn't take a weekend shift unless there was a strong monetary incentive. Another critical aspect I considered was Clipboard's revenue and how this varies by day of the week. I wanted to understand whether revenue patterns were consistent or if certain days were significantly more profitable or problematic. Based on these initial thoughts, I decided to explore these questions further, looking for trends and connections that could help explain the key factors driving revenue loss and unclaimed shifts.

Total Estimated Lost Revenue (by Slots): 697745.40					
Revenue Loss by Day & Slot (Based on Unclaimed Shifts):					
	DAY_OF_WEEK	SLOT	NUM_UNCLAIMED_SHIFTS	TOTAL_UNCLAIMED_HOURS	LOST_REVENUE
0	Friday	am	710	6092	39876.214916
1	Friday	noc	387	3203	20443.796757
2	Friday	pm	973	7974	53999.547219
3	Monday	am	759	6441	36941.590921
4	Monday	noc	523	4360	27655.506212
5	Monday	pm	785	6547	41077.760366
6	Saturday	am	881	7633	47412.180673
7	Saturday	noc	544	4528	33052.107342
8	Saturday	pm	1216	9989	68485.380465
9	Sunday	am	1047	8972	50801.726336
10	Sunday	noc	653	5477	32270.688429
11	Sunday	pm	931	7702	47096.568445
12	Thursday	am	509	4381	24851.359979
13	Thursday	noc	312	2645	14368.243369
14	Thursday	pm	779	6417	33671.783057
15	Tuesday	am	555	4818	30523.799322
16	Tuesday	noc	401	3384	24395.660858
17	Tuesday	pm	714	5911	20683.640433
18	Wednesday	am	535	4563	13550.670000
19	Wednesday	noc	388	3299	14490.652709
20	Wednesday	pm	646	5315	22096.520064

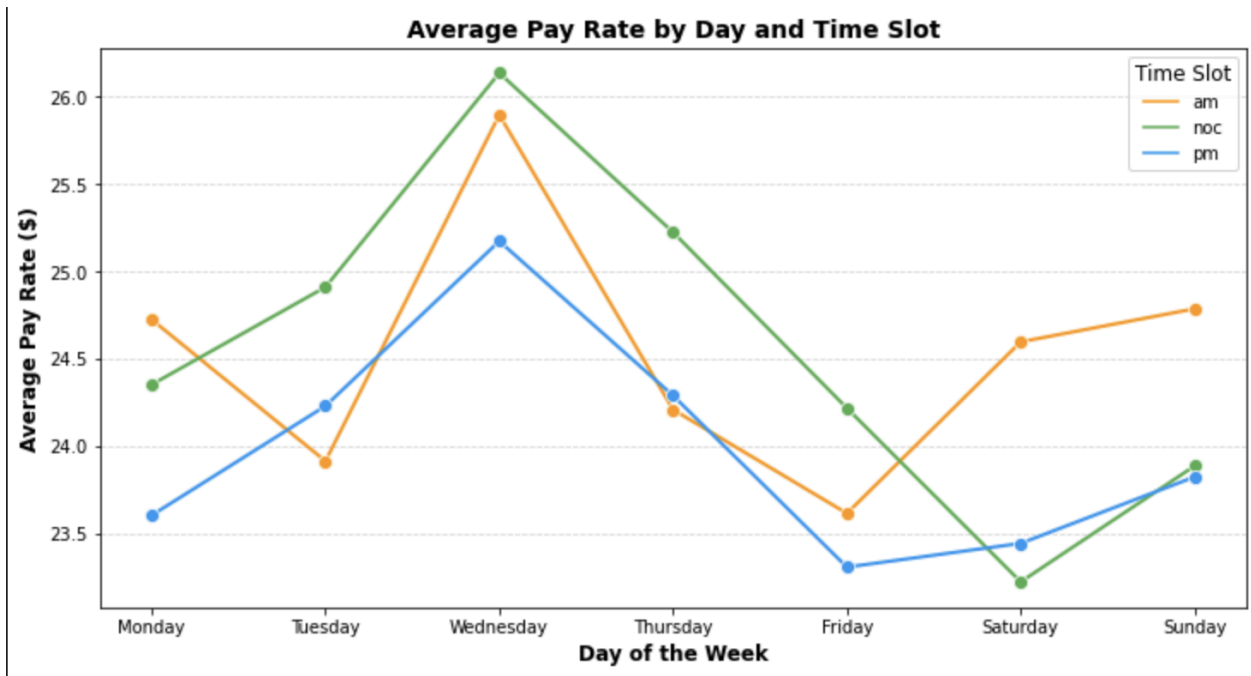
Revenue Loss Breakdown by Day and Slot:

To start my analysis, I explored unclaimed shift offers, specifically looking at which shifts were never claimed, what day of the week they occurred on, and their respective time slots. My goal was to identify patterns in unclaimed shifts and connect them to my initial question about whether workers preferred working on weekends.

A clear pattern emerged: weekend shifts had significantly higher numbers of unclaimed shifts compared to weekdays. Friday night, Saturday morning and night, and Sunday morning and night had the highest number of unclaimed shifts. This makes sense, as weekends are when most people prefer to spend time with family, take trips, or engage in personal activities, making these shifts less desirable.

This pattern has major financial consequences. The total estimated lost revenue from unclaimed shifts from September 2024 to January 2025 was approximately \$697,745.40, with the highest losses occurring on Saturday AM (\$47,412.18), Saturday PM (\$68,485.38), Friday PM (\$53,999.55), and Sunday PM (\$47,096.57). These shifts consistently have high numbers of unclaimed hours, making them a significant source of revenue loss for Clipboard.

For each unique shift that remained unclaimed despite being offered, I calculated lost revenue by multiplying its duration (in hours) by the average difference between the charge rate and pay rate for claimed shifts within the same day and time slot. Using the claimed shift rates provides a more accurate estimate, as the actual revenue generated is lower than the difference between the charge and pay rate for unclaimed shift offers. This calculation was applied to all unclaimed shifts and then aggregated across different time slots and days of the week.



Pay Rate Analysis: Weekend Shifts vs Weekdays:

As shown in the graph, weekend shift offers tend to have lower pay rates compared to weekdays, with Wednesday standing out as the highest-paying day across all shifts. The lower pay rates on weekends might be contributing to the higher number of unclaimed shifts, as workers may be less motivated to take them on. If weekend pay rates were adjusted to be more competitive over other days, it could incentivize workers to accept these shifts, thereby reducing unclaimed shifts and increasing overall revenue for the

company. One thing that I would like to know more about would be why Wednesday has such higher rates for all slots of the shift compared to others.

Average Pay Rate by Day & Slot with Percent Difference:					
	DAY_OF_WEEK	SLOT	CLAIMED_AVG_PAY_RATE	UNCLAIMED_AVG_PAY_RATE	PERCENT_DIFF
0	Friday	am	25.760351	23.524743	9.503219
1	Friday	noc	25.695921	24.118655	6.539611
2	Friday	pm	25.407823	23.233102	9.360443
3	Monday	am	26.404187	24.638873	7.164751
4	Monday	noc	26.023727	24.261133	7.265094
5	Monday	pm	25.955221	23.444959	10.707043
6	Saturday	am	26.387001	24.513246	7.643849
7	Saturday	noc	24.635105	23.160981	6.364689
8	Saturday	pm	25.578029	23.369157	9.452080
9	Sunday	am	27.016627	24.660950	9.552257
10	Sunday	noc	26.608740	23.732038	12.121600
11	Sunday	pm	26.411833	23.675219	11.558980
12	Thursday	am	26.909059	24.085442	11.723333
13	Thursday	noc	27.199072	25.014371	8.733784
14	Thursday	pm	26.754178	24.166676	10.706897
15	Tuesday	am	26.158041	23.806333	9.878498
16	Tuesday	noc	25.366501	24.872088	1.987825
17	Tuesday	pm	28.367788	23.974779	18.323461
18	Wednesday	am	29.604280	25.684241	15.262430
19	Wednesday	noc	27.991309	25.965245	7.802986
20	Wednesday	pm	28.035287	24.986130	12.203396

After analyzing lost revenue from unclaimed shifts and looking at no real monetary incentive for weekend shifts, I wanted to explore how pay rates differ between claimed and unclaimed shifts and whether higher pay could explain shifts being claimed more often.

Across all days and time slots, the claimed average pay rate is consistently higher than the unclaimed average pay rate.

Interestingly, some weekday shifts (like Wednesday AM) have a much larger pay difference than weekend shifts.

- Wednesday AM: Claimed shift avg pay = \$29.60, Unclaimed shift offer avg pay = \$25.68 → 15.26% difference
- Tuesday PM: Claimed shift avg pay = \$28.37, Unclaimed shift offer avg pay = \$23.97 → 18.32% difference

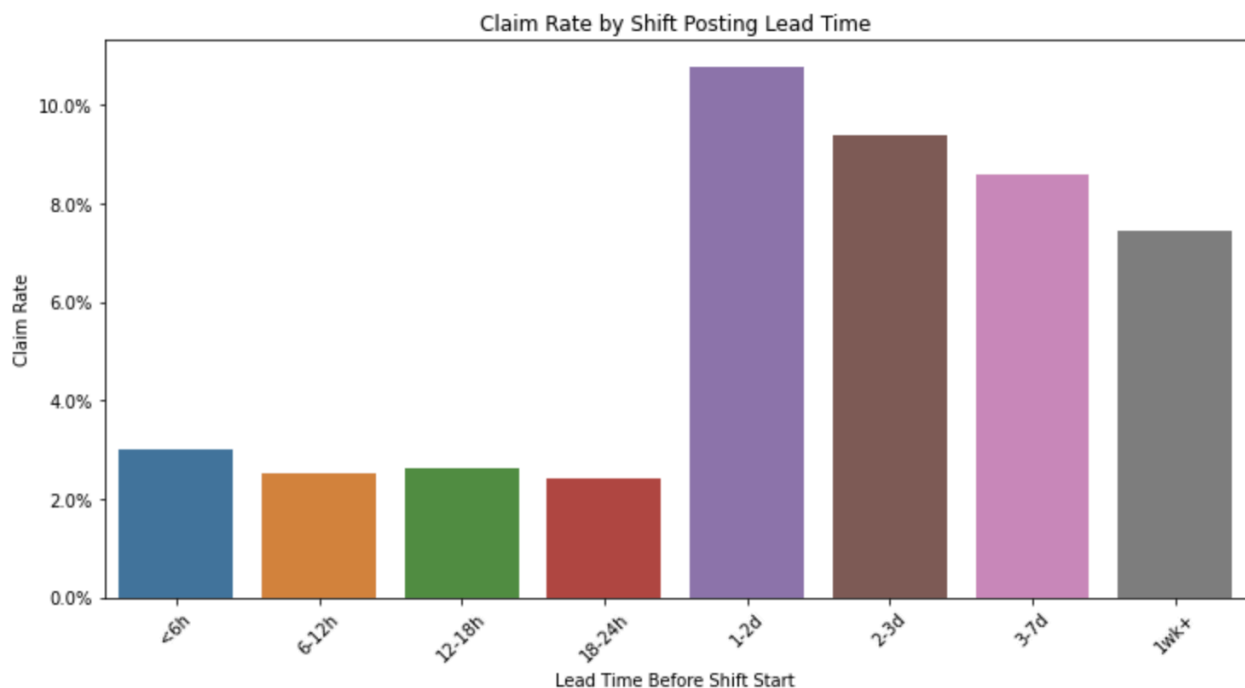
This could indicate that when shifts must be covered, employers are more willing to pay higher rates, while weekend shifts remain relatively under-incentivized despite higher unclaimed rates. I find this intriguing and would like to learn more about this by talking to healthcare workers and workplaces as to why that is the case.

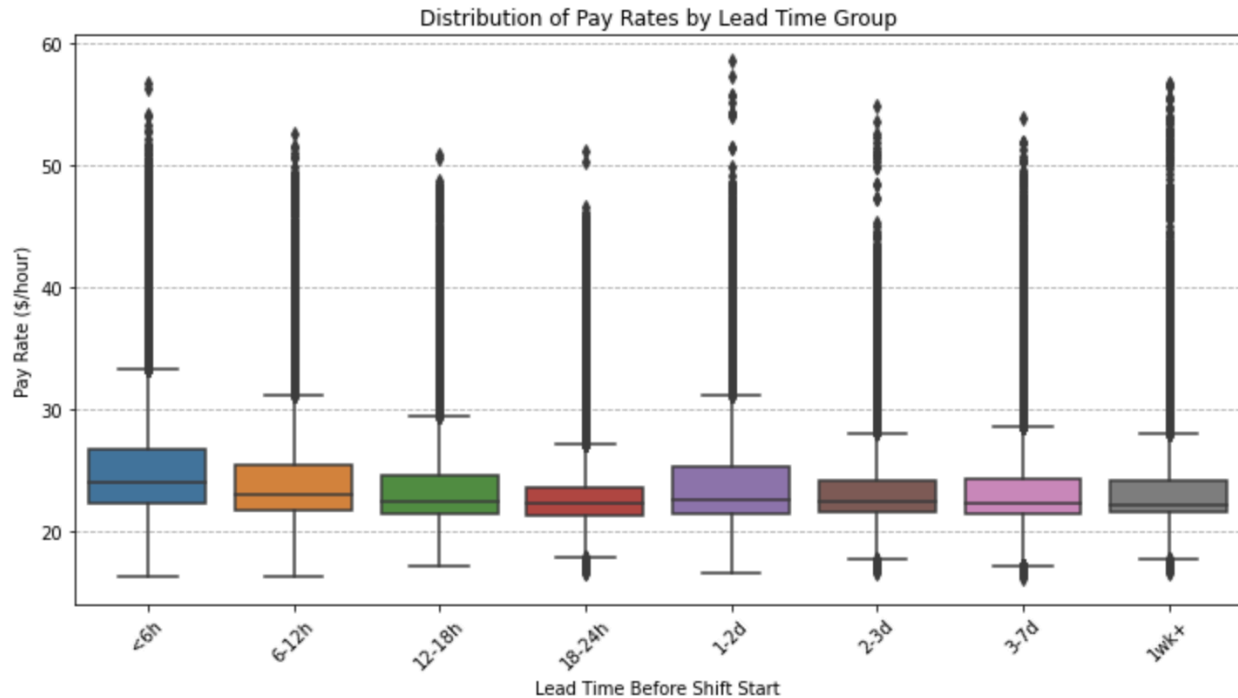
Revenue Implications and Strategic Recommendations:

Looking at the overall revenue, the company's expected revenue from unclaimed shifts is generally higher than the actual revenue from claimed shifts. This indicates that the company might be setting its pay rates too low for those particular shifts, leading to more unclaimed shifts. One possible strategy to address this could be to lower the expected revenue per shift (i.e., reduce the difference between charge rates and pay rates). While this would decrease the margin per shift, it could lead to more shifts being claimed, ultimately resulting in higher overall revenue due to increased shift coverage.

By focusing on balancing pay rates across shifts and adjusting weekend incentives, the company could improve revenue generation and reduce the financial impact of unclaimed shifts.

Analysis of Shift Posting Times and Their Impact on Claim Rates:





To further explore reasons behind unclaimed shifts, I looked into the relationship between lead time—the time between when the shift is posted and its start time—and claim rates which represent the percentage of shifts claimed out of the total offers available. The data revealed that as the lead time decreased, the claim rate also dropped. For example:

- Shifts posted with less than 6 hours' lead time had a claim rate of 3.0%.
- Shifts posted 1-2 days in advance had a claim rate of more than 10%.

This suggests that workers may be less willing to accept shifts posted closer to the start time, likely due to lack of time to prepare, scheduling conflicts, or personal reasons. Therefore, posting shifts too close to the start time could be a significant contributor to the unclaimed shifts issue as well.

Pay Rates and Financial Incentives:

I also examined the pay rates for shifts based on their lead time group to determine if financial incentives for last-minute shifts were sufficient. The data shows that pay rates remain relatively consistent across lead times:

- Shifts posted with less than 6 hours' lead time had an avg pay rate of \$25.14.
- Shifts posted 6-12 hours in advance had an avg pay rate of \$24.14.
- Shifts posted 12-18 hours in advance had an avg pay rate of \$23.61.
- Shifts posted more than a week in advance had an avg pay rate of \$23.98.

There is a very small increase in pay for shifts posted closer to the start time, suggesting that the financial incentive for workers to take last-minute shifts is insufficient.

Given the findings, adjusting the pay or charge rates for shifts posted within shorter lead times could be beneficial. Offering a significantly higher pay rate for shifts posted 6 hours or less in advance would provide stronger incentives for workers to accept these shifts, reducing the number of unclaimed shifts. This strategy would not only compensate workers more fairly but also help the company fill more shifts, improving overall revenue.

Extra: Exploring No-Shows

	SLOT	NUM_NCNS
0	am	17
1	noc	6
2	pm	8

Lastly, I wanted to explore the issue of no-shows to identify any patterns related to low pay rates, late shift postings, or specific time slots. One key finding, 17 out of 31 no-shows were for AM shifts. The reasons behind this could include factors like fatigue, personal appointments, or simply oversleeping, making early morning shifts less appealing. To address this, further investigation into the causes of these no-shows is needed. Additionally, implementing reminder or confirmation protocols for AM shifts could potentially help reduce this and improve shift coverage.